that these visualizations effectively and accurately communicate experimental results. Second, during the manuscript writing reflection stage, VLMs evaluate figures and their captions, enhancing the visual clarity and coherence of the resulting paper.

In the paper-writing process, we extract screenshots of figures alongside their captions and the corresponding text from the paper that references them (identified by the keyword "Figure X"). These images and textual references are then provided to the VLM, which performs multiple quality checks, including verifying the alignment between figures and captions, identifying issues with visual clarity (e.g., missing legends, unclear labels), and detecting potential duplication of figures in the main text and appendix. Through the iterative integration of VLM feedback, we significantly enhance the visual quality and clarity of manuscripts generated by THE AI SCIENTIST-V2.

4. Human Evaluation of Manuscripts Generated by THE AI SCIENTIST-V2

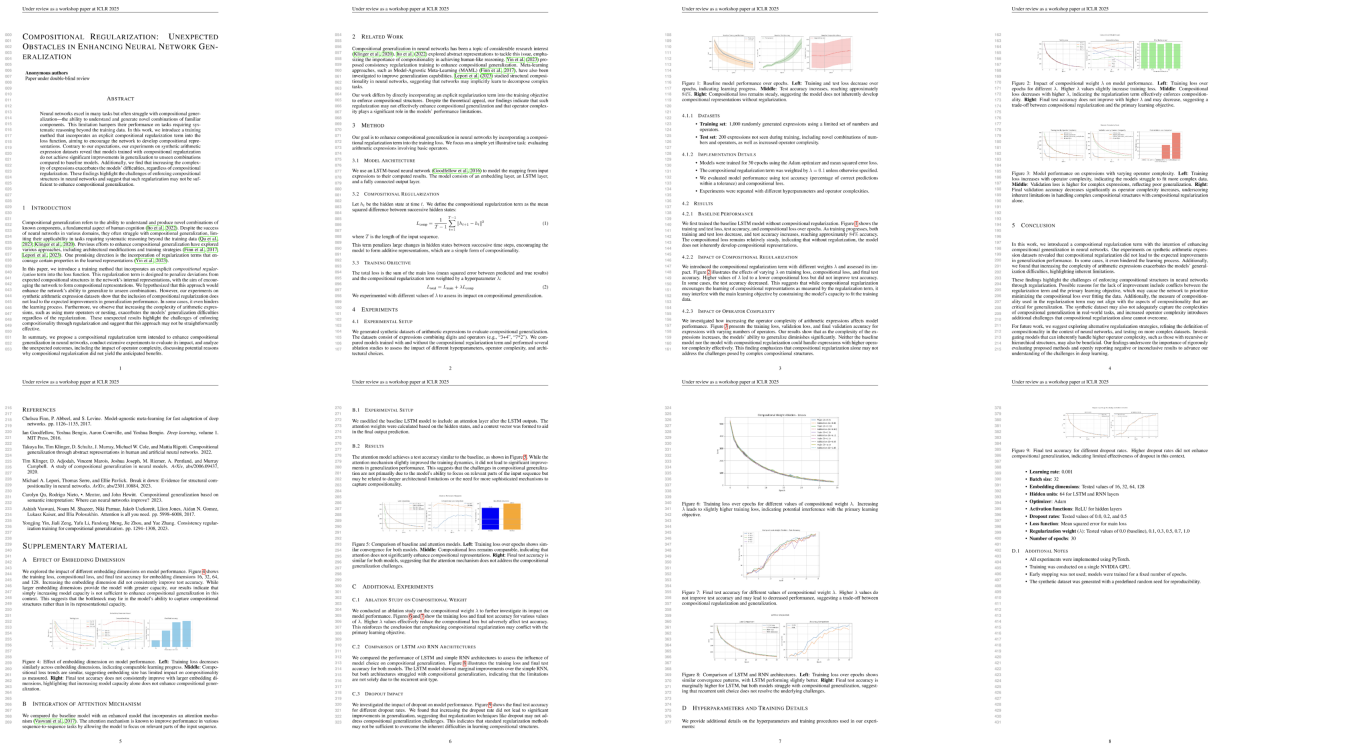

Figure 3 | **Peer-reviewed ICBINB workshop paper generated by THE AI SCIENTIST-V2.** The paper investigates the usage of a temporal consistency regularizer on the embeddings of an LSTM-based sequence model. The results discuss the effect of the regularizer on compositional regularization and highlight the difficulty of training models capable of improved generalization. It received peer-review scores of 6 (weak accept), 7 (accept), and 6 (weak accept) before meta-review and ranked among the top 45% submitted workshop papers.

To rigorously evaluate the capabilities and limitations of our automated scientific discovery system, we conducted a human evaluation study in collaboration with the organizers of the ICLR 2025 workshop, "I Can't Believe It's Not Better" (ICBINB). As detailed below, our evaluation included submitting fully automated manuscripts generated by THE AI SCIENTIST-V2 to the official peer-review process of the workshop.

4.1. Evaluation Methodology

Our evaluation process involved the following carefully controlled steps: